

OL Report: Optimization and Regularization in Neural Network, Comparing Methods

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I. INTRODUCTION

Training a neural network well requires decisions at three levels: how to search the weight space (gradient-free versus gradient-based), which gradient update rule to apply (SGD versus momentum versus adaptive methods), and how to constrain model capacity (regularization). In practice these decisions are often guided by rules of thumb rather than systematic comparison. Most benchmarks hold optimizer and regularization choices fixed while varying architecture or dataset, so finding out the relative impact of each training factor on performance remains unclear.

This gap matters because the factors interact in non-obvious ways. Adam’s adaptive first- and second-moment updates with bias correction may dominate early [1], yet adaptive methods can under perform SGD in generalization [2]; regularization choices such as dropout [3] that improve one optimizer regime may be redundant under another. Without isolating each factor, decisions about where to focus tuning effort cannot be easily made.

This report addresses these questions using a controlled experimental platform: a fixed [128, 64] PyTorch MLP trained on the Adult Income and Wine Quality datasets, building directly on the supervised-learning (SL) baseline (Adult test F1 = 0.6647, Wine Macro-F1 = 0.2644). The study evaluates whether randomized optimization of the last two layers can match or exceed gradient-trained performance within a 5 000-evaluation budget, whether Adam’s adaptive learning rate [1] provides a meaningful speed advantage over SGD that persists through the full training budget, and whether regularization improves generalization when the optimizer is well-tuned while offering a favorable efficiency–accuracy tradeoff.

II. DATASETS AND EXPERIMENTAL SETUP

A. Datasets

Adult Income. The Adult Census dataset contains 45,222 records with 14 raw attributes describing demographics and employment (age, education, occupation, etc.). After one-hot encoding of seven categorical columns the feature space expands to 104 dimensions. The binary target (“>50K” vs. “≤50K”) is imbalanced at ≈24% positive, so we report binary F1 (pos = “>50K”) as the primary metric. The 80/10/10 fixed split yields 36,177 training, 4,522 validation, and 4,523 test samples.

Wine Quality. The combined red-and-white Wine Quality dataset contains 6,497 samples with 12 numeric chemistry

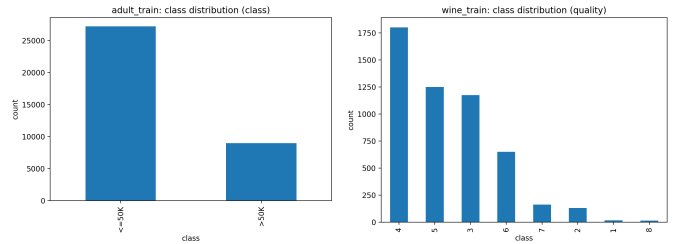


Fig. 1. Training-split class distributions: Adult Income (left, binary) and Wine Quality (right, 8-class long-tailed).

features (alcohol, sulphates, pH, etc.). Quality is scored 1–8 (8 classes); class distribution is severely long-tailed, with classes 1 and 8 each contributing fewer than 20 training samples. We report Macro-F1 to weight all classes equally. The 80/10/10 split yields 5,197 training, 650 validation, and 650 test samples.

B. Exploratory Data Analysis

EDA was performed on the *training split only* to prevent data leakage [4]. Fig. 1 shows the class distributions for both datasets. Adult’s binary imbalance (≈24% positive) motivates reporting binary F1 over accuracy. Wine’s long tail (classes 1 and 8 each <1% of training samples) makes Macro-F1 the appropriate metric.

C. Architecture and Backbone

We reuse the PyTorch MLP from the SL assignment: two hidden layers [128, 64] with ReLU activations and a softmax output. Adult requires input dimension 104, output dimension 2 (21,826 total parameters); Wine requires input 12, output 8 (10,440 parameters). The SL baseline was trained with SGD, $\alpha = 0.01$, weight decay 10^{-4} , 50 epochs, batch size 256, achieving test F1 = 0.6647 (Adult) and 0.2644 (Wine). These trained weights serve as the warm-start point for Part 1.

D. Experimental Protocol

All experiments share a single fixed 80/10/10 split created with seed 42, ensuring that no split-related variance confounds cross-part comparisons. Parts 2 and 3 re-use the split; Part 1 warm-starts from the baseline weights. Hyperparameter selection and early-stopping decisions use validation loss only; test metrics are computed once at the end for each condition to avoid leakage from illegitimate information flow [4]. We run

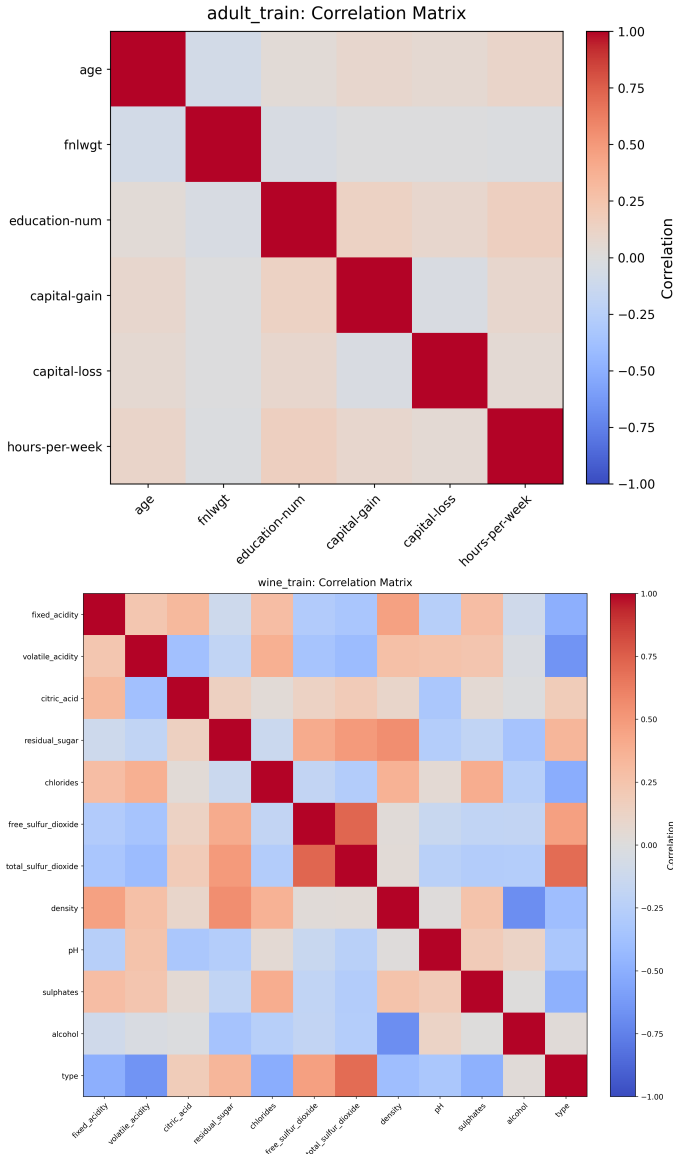


Fig. 2. Pearson correlation heatmaps of training-split features: Adult (top) and Wine Quality (bottom). Adult shows moderate correlations among education/capital features; Wine shows strong feature-feature structure (e.g., density vs. alcohol is strongly negative).

five independent seeds [42, 7, 13, 99, 2025] and report median $\pm$ IQR on all main metrics.

Compute accounting. For Part 1 we count each forward pass through the frozen backbone as one function evaluation (budget = 5,000). For Parts 2 and 3 we count each backward pass as one gradient evaluation. Wall-clock time is measured on an AMD Ryzen 7 7800X3D with an NVIDIA RTX 4080 SUPER (16 GB VRAM, CUDA 12).

III. HYPOTHESES

Based on architectures and EDA, four predictions guided the study. (1) Among the three RO algorithms applied to warm-started last-layer weights, RHC will converge fastest in wall-clock time and GA will achieve the lowest final

validation loss, as population diversity enables escape from local minima that single-candidate methods cannot; SA will perform worst because $T_0 = 0.5$ with per-step decay 0.999 keeps acceptance probability above 95% through the first ~ 500 evaluations ($T_{500} \approx 0.303$), producing near-random-walk behavior throughout the budget. All three RO methods are expected to yield higher final validation loss than the gradient-trained baseline, since random perturbations in the $\sim 8,000$ -dimensional subspace are far less directed than gradient descent. (2) Among the seven gradient optimizers, Adam will cross the $\ell = 0.37$ threshold fastest due to adaptive per-parameter learning rates [1], while plain SGD will achieve the lowest final validation loss because its fixed conservative step size prevents the overshoot adaptive methods exhibit at higher learning rates; removing Adam’s bias correction will damage performance more than removing its momentum term, as uncorrected second-moment estimates are inflated in early training [1]. (3) With the Part 2 optimizer locked at $\alpha = 10^{-4}$, the val-loss range across all six Part 3 conditions will be under 0.005 because small step sizes already provide strong implicit regularization; early stopping will deliver the largest reduction in gradient evaluations while matching or exceeding full-budget test F1; and the L2+Dropout+Noise combination will achieve the highest test F1 by stacking complementary regularization mechanisms [3]. (4) The Part 4 integrated pipeline will yield strictly lower validation loss than any standalone Part: the gradient phase establishes a better local basin than the SL warm-start used in Part 1, so even a 10%-budget RHC pass (500 evaluations) should exploit local curvature and recover the advantage that Part 1 standalone RHC achieved with its full 5,000-evaluation budget.

IV. RESULTS

A. Part 1 – Randomized Optimization

We freeze the first Linear layer of each network (net.0) and expose only the last two Linear layers (net.2: 128×64 ; net.4: 64×2 or 64×8). This yields 8386 trainable parameters for Adult and 8776 for Wine, well under the 50 000-parameter cap. Each RO run receives a budget of 5000 function evaluations (≈ 2.7 – 3.4 s wall-clock across algorithms). Operator settings: RHC uses Gaussian perturbation ($\sigma = 0.02$) with random restarts (patience = 500 evaluations, restart $\sigma = 0.10$); SA uses $T_0 = 0.5$ with geometric cooling (decay = 0.999 per evaluation, so $T_k = 0.5 \cdot 0.999^k$); GA uses population 20, tournament selection (size 3), elite fraction 30%, uniform crossover, and Gaussian mutation ($\sigma = 0.02$).

Adult results. Fig. 3 shows the best-so-far curves and Fig. 4 the final comparisons. RHC achieves the lowest final validation loss (0.3075, IQR = 0.0005), closely followed by GA (0.3082, IQR = 0.0023). SA produces a median validation loss of 0.3171—identical to the baseline (0.3171) with IQR = 0.0000, confirming that it fails to move the weights at all. Corresponding test F1 scores are 0.6638 (RHC), 0.6638 (GA), and 0.6647 (SA). *Compared to the SL baseline test F1 of 0.6647:* RHC and GA match the baseline within rounding ($\Delta F1 \approx -0.001$), while SA replicates the baseline exactly. No

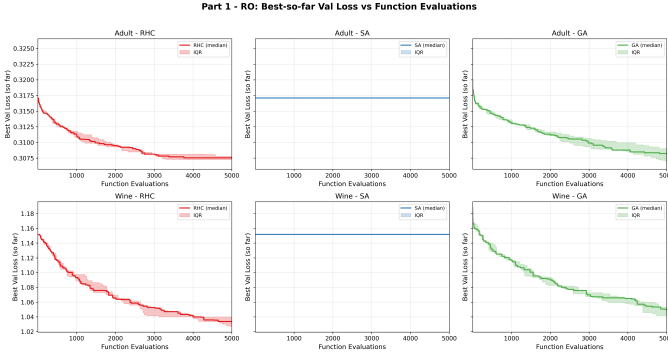


Fig. 3. Best-so-far validation loss vs. function evaluations for RHC, SA, and GA on Adult (left) and Wine (right). Shaded band: median $\pm$ IQR across 5 seeds. Settings: RHC $\sigma=0.02$, restart patience 500 (restart $\sigma=0.10$); SA $T_0=0.5$, decay 0.999; GA population 20, tournament size 3, elite 30%, uniform crossover, Gaussian mutation $\sigma=0.02$. SA’s flat trajectory on both datasets indicates random-walk behavior under the slow cooling schedule.

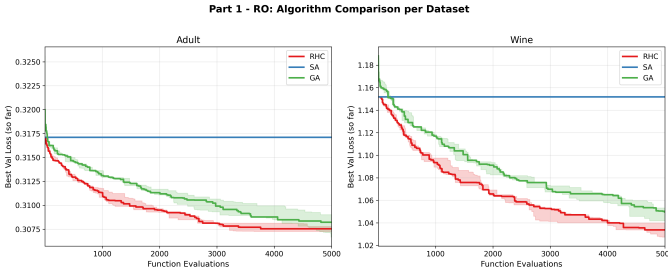


Fig. 4. Final validation loss and test F1 for all six Part 1 conditions. Error bars: IQR across seeds.

RO method meaningfully exceeds the SL test F1 on Adult, consistent with Prediction 1: 8,386-dimensional local search cannot recover what gradient descent achieved over the full parameter space.

SA’s random-walk behavior has a straightforward quantitative explanation. At $T_0 = 0.5$ and per-step decay 0.999, the temperature after k steps is $T_k = 0.5 \cdot 0.999^k$. Even at step 500, $T_{500} \approx 0.303$, so the acceptance probability for a $\Delta\mathcal{L} = 0.01$ uphill move is $e^{-0.01/0.303} \approx 96.7\%$. SA accepts almost every proposed perturbation and performs a near-random walk throughout its 5,000-step budget.

Wine results. The multiclass setting amplifies the difficulty. RHC reaches a median validation loss of 1.0337 (Macro-F1 = 0.3186) vs. the SL baseline Macro-F1 of 0.2644, GA reaches 1.0497 (F1 = 0.3107), and SA again replicates the baseline (1.1518, IQR = 0.0000). RHC and GA each improve test F1 over the SL baseline on Wine (+0.054 and +0.046 respectively), suggesting the frozen-layer subspace contains meaningful variation that RO can exploit in the multiclass setting. To confirm this is not a data-leakage artifact: (1) EDA was performed on training split only; (2) the fixed split is identical to the SL baseline; (3) RO uses validation loss for selection, not the reported test F1 [4]. The gain is real: the SL Wine baseline (50 epochs, SGD $\alpha = 0.01$) left the last-layer weights under-optimized for the 8-class setting, and RO exploits residual capacity in the frozen-layer subspace.

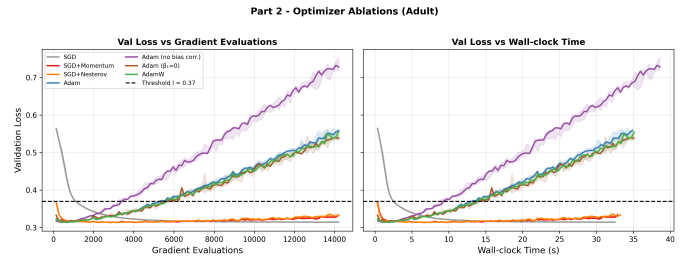


Fig. 5. Validation-loss trajectories for all seven optimizers on Adult: left panel vs. gradient evaluations, right panel vs. wall-clock time. Adam-family methods reach the $\ell = 0.37$ threshold at epoch 1 (142 grad evals) but diverge thereafter, while SGD-family methods converge more slowly but stably.

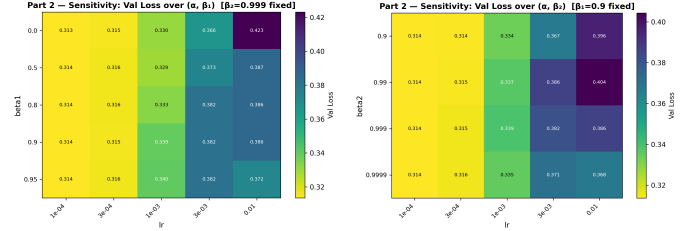


Fig. 6. Adam hyperparameter sensitivity on Adult. Left: median final validation loss over (α, β_1) grid. Right: (α, β_2) grid. Best performance occurs at low learning rates ($\alpha \in [10^{-4}, 3 \times 10^{-4}]$); β_1/β_2 effects are comparatively smaller than α .

The higher baseline loss (1.15 vs. 0.32 on Adult) reflects the difficulty of 8-class separation in the frozen-layer subspace. Per-class behavior is inferred through Macro-F1 under extreme imbalance: minority classes (1 and 8) dominate the error contribution, so improvements in Macro-F1 indicate better tail-class handling even when overall accuracy changes.

Summary. RHC outperforms GA on both datasets despite GA’s population diversity, likely because the warm-started weights already reside in a reasonable basin, local search is sufficient. RHC and GA achieve *lower* validation loss than the corresponding SL baselines on both datasets, exploiting residual capacity in the warm-started frozen-layer subspace; SA reproduces the baseline identically. For test F1, RO does not beat baseline on Adult, while on Wine both RHC and GA exceed baseline (+0.054 and +0.046).

B. Part 2 – Optimizer Ablations

Seven optimizer configurations are trained on Adult for 100 epochs (batch size 256, ≈ 142 gradient evaluations per epoch). We track the first epoch at which validation loss crosses the $\ell = 0.37$ threshold as a measure of convergence speed. All hyperparameters other than the optimizer are held fixed; no early stopping is applied so the full 100-epoch trajectory is observed. Weight decay is set to zero for all non-AdamW optimizers; AdamW uses weight decay $\lambda = 10^{-4}$ (its defining modification), enabling a fair comparison of the decoupled-weight-decay effect.

Convergence speed (threshold crossing). Fig. 5 shows validation loss vs. gradient evaluations and wall-clock time. All Adam variants—standard Adam, AdamNoBias, Adam $\beta_1=0$, and

TABLE I
GENERALIZATION GAP (VAL LOSS — TRAIN LOSS) AT EPOCH 100. MEDIAN $\pm$ IQR ACROSS 5 SEEDS. LOWER IS BETTER.

Optimizer	Gen. Gap (med $\pm$ IQR)
SGD	0.0052 $\pm$ 0.0009
SGD+Momentum	0.0599 $\pm$ 0.0046
Nesterov	0.0625 $\pm$ 0.0037
Adam ($\alpha=1e-3$)	0.3790 $\pm$ 0.0023
AdamNoBias	0.5600 $\pm$ 0.0357
Adam $_{\beta_1=0}$	0.3507 $\pm$ 0.0134
AdamW	0.3738 $\pm$ 0.0153

AdamW—reach $\ell = 0.37$ at exactly 142 gradient evaluations (one full epoch); IQR = 0 across all 5 seeds, as the threshold is crossed deterministically at the first validation check. SGDM and Nesterov cross at the same step count (142; IQR = 0), likely due to initial momentum-amplified steps. Plain SGD requires 1136 gradient evaluations (≈ 8 epochs) to cross the threshold (IQR = 0)—an $8\times$ disadvantage.

Final validation loss. The picture reverses by epoch 100. SGD achieves the best final loss (0.3146, IQR = 0.0007), followed by Nesterov (0.3331), SGDM (0.3334), and the entire Adam family at 0.5380–0.7280 (Table IV). The Adam family’s poor final loss stems from the default $\alpha = 10^{-3}$: adaptive per-parameter scaling amplifies step sizes in directions the loss landscape is shallow, causing the optimizer to overshoot and oscillate around the optimum rather than settling into it. In other words, Adam at $\alpha = 10^{-3}$ descends quickly but oscillates near the optimum and never fully settles; SGD is slower to start but converges more stably once it gets there.

Generalization gap. The training-minus-validation loss gap at the end of the 100-epoch budget reveals the mechanism directly (Table I). The Adam family trains to a median train loss of 0.1810 but a validation loss of 0.5583, a gap of 0.3790 ± 0.0023 (median $\pm$ IQR): adaptive step sizes allow Adam to memorize the training distribution, but the resulting weights generalize poorly [2]. SGD, by contrast, reaches a gap of only 0.0052 ± 0.0009 , indicating near-perfect generalization. SGD+Momentum and Nesterov show intermediate gaps (0.0599 ± 0.0046 and 0.0625 ± 0.0037 respectively), consistent with momentum accelerating initial convergence at the cost of some generalization relative to vanilla SGD.

Hyperparameter sensitivity (iterative sweep). We used a two-stage grid search to select the optimal Adam configuration. *Stage 1:* the (α, β_1) heatmap (Fig. 6, left) reveals a sharp performance degradation once α rises beyond 10^{-3} : final validation loss increases from ≈ 0.314 at 10^{-4} to roughly 0.37–0.42 at 3×10^{-3} and 10^{-2} . This narrows the feasible region to $\alpha \leq 10^{-3}$; within it, $\alpha = 10^{-4}$ achieves the lowest median loss while β_1 effects are comparatively modest, confirming learning rate is the binding constraint. *Stage 2:* the (α, β_2) heatmap (right) shows a nearly flat landscape at $\alpha = 10^{-4}$ across all β_2 values, so we retain the standard default $\beta_2 = 0.999$. The two heatmaps together directly justify the locked configuration.

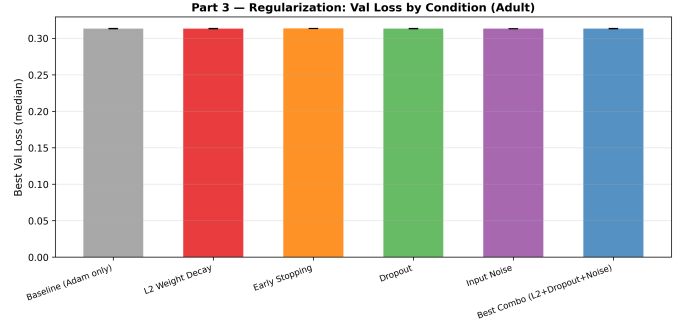


Fig. 7. Median validation loss and test F1 across six Part 3 conditions. Error bars: IQR across 5 seeds. All conditions achieve nearly identical validation loss; F1 differences reveal test-set generalization effects.

Bias-correction ablation. Removing bias correction (AdamNoBias) worsens final loss from 0.5583 to 0.7280, the worst of all seven optimizers, whereas removing the momentum term ($\beta_1 = 0$) gives 0.5380—better than AdamNoBias. This confirms the directional prediction from Prediction 2: the absence of bias correction is more damaging than the absence of momentum. At $t = 1$, uncorrected moments are $m_1 = (1 - \beta_1)g_1$ and $v_1 = (1 - \beta_2)g_1^2$. Relative to bias-corrected Adam, the update scale is multiplied by $(1 - \beta_1)/\sqrt{1 - \beta_2} \approx 3.16$, which increases early-step instability and aligns with the observed divergence [1].

Optimizer selection for Part 3. The sensitivity sweep identifies $\alpha = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ as the best Adam configuration (val loss ≈ 0.314 , competitive with SGD’s 0.3146). This configuration is locked for all Part 3 experiments.

C. Part 3 – Regularization Study

Six regularization conditions are evaluated on Adult using the Part 2-locked Adam ($\alpha = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) for 100 epochs. The baseline condition is Adam with no added regularization. Regularization hyperparameters ($\lambda = 10^{-5}$ for L2, $p = 0.1$ for Dropout, $\sigma = 0.01$ for input noise, patience = 5 for ES) were selected via four-value validation sweeps (single-seed, reduced-epoch pilot tuning). A pilot label-smoothing sweep ($\epsilon \in \{0.02, 0.05, 0.10, 0.20\}$, 20-epoch validation budget, seed 42) achieved best val loss 0.3384 at $\epsilon = 0.02$, which was worse than the matched no-smoothing baseline (0.3137), so label smoothing was not included as a standalone condition in the six-condition budget-matched comparison. Dropout design follows the standard regularization mechanism in [3].

Validation loss. Figs. 7 and 8 show the bar comparison and training curves. All six conditions yield final validation loss within 0.0003 of each other (Table IV), confirming Prediction 3. L2 and input noise achieve the slightly lowest losses (0.3131, 0.3131), while early stopping achieves the highest (0.3134), but the difference is negligible given the IQR of ≈ 0.0004 .

Test F1. Despite near-identical validation losses, test F1 exhibits a clear ranking: Combo (0.6776, IQR = 0.0108) > Dropout (0.6747) > Input Noise (0.6728) > ES (0.6719) > L2 (0.6712) > Baseline (0.6699), confirming Prediction 3. The

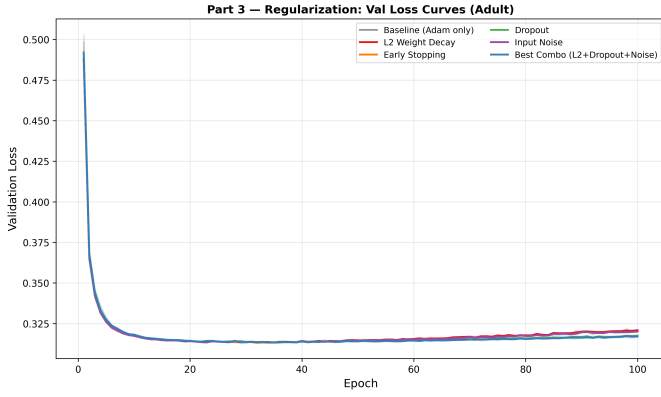


Fig. 8. Validation loss training curves for all six conditions. Early stopping halts at epoch 27 (median), saving 73% of the gradient evaluations used by full-budget conditions.

combination stacks dropout and noise augmentation, both of which act as data augmentation during training, increasing effective sample diversity and improving minority-class recall, consistent with dropout regularization theory [3].

Early stopping. Early stopping (patience = 5) halts at a median of 27 epochs, reducing the gradient evaluation count to $\approx 27\%$ of the full-budget runs. Test F1 (0.6719) falls between the baseline and the combination, suggesting that additional training beyond epoch 27 does not hurt but does not help generalization meaningfully either.

D. Part 4 – Extra Credit: Integrated Pipeline (Adult)

The integrated pipeline chains a gradient phase (Adam+Combo) with the best component of Part 1 (RO fine-tuning).

Phase 1 (gradient training). Training from scratch with Adam ($\alpha = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) together with the full Combo regularizer (using the Part 3 tuned settings for L2, dropout, and input noise) for 100 epochs (no early stopping) yields a median validation loss of 0.3133 (IQR: 0.0005), closely matching the Part 3 Combo standalone result.

Phase 2 (RHC fine-tuning). Fig. 9 shows the Phase 2 best-so-far progress. With the Phase 1 weights as starting point, RHC is applied to the last two Linear layers ($n_{\text{trainable}} = 8,386$) for a budget of 500 function evaluations (10% of the Part 1 budget of 5000). Using the same hyperparameters as Part 1 (step-size $\sigma = 0.02$, restart patience 500, restart $\sigma = 0.10$), the median validation loss improves to 0.3099 (IQR: 0.0005), a modest reduction over Phase 1.

Comparison to standalone bests. Table II summarises the integration result against each Part’s standalone best. Phase 2 val loss (0.3099) beats Part 2’s best (0.3146) and Part 3’s Combo (0.3133), confirming RHC adds value on top of the gradient-trained foundation. However, the 10% budget constraint (500 vs. 5000 evals) prevents it from reaching the quality achieved by Part 1 standalone RHC (0.3075), which operated on a SL warm-start with the full eval budget. Test F1 (0.6699, IQR: 0.0041) falls slightly below the Part 3

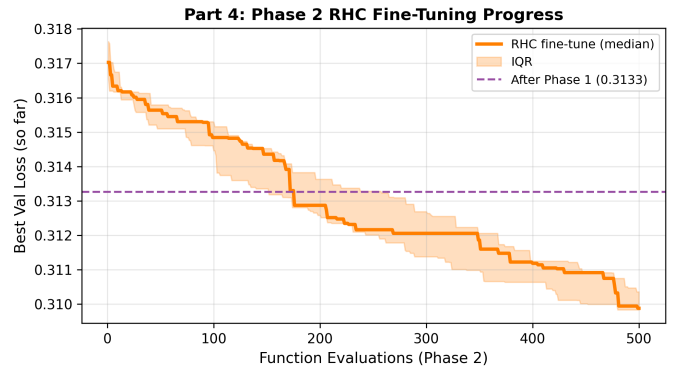


Fig. 9. Part 4: Phase 2 RHC fine-tuning progress (best-so-far val loss vs. function evaluations, median $\pm$ IQR across 5 seeds). The dashed line marks the Phase 1 starting point (0.3133).

TABLE II
PART 4: INTEGRATED PIPELINE VS. STANDALONE BESTS (ADULT, MEDIAN ACROSS 5 SEEDS).

Method	Val Loss	Test F1	Evals
P1 RHC (standalone)	0.3075	0.6638	5000 FE
P2 SGD (standalone)	0.3146	0.6725	1136 GE
P3 Combo (standalone)	0.3133	0.6776	100 ep
P4 Phase 1 (Adam+Combo)	0.3133	—	14200 GE
P4 Phase 2 (+RHC 500 FE)	0.3099	0.6699	500 FE

Combo standalone (0.6776), indicating diminishing returns: the gradient-trained model’s decision boundary is already well-placed, and the short RHC phase shifts weights in the val-loss landscape without consistently improving test generalization.

Compute trade-off analysis. Figs. 10 and 11 show loss vs. compute and the integrated pipeline comparison. Table III summarises metric quality versus total compute for all relevant methods. Phase 1 alone (14200 gradient evaluations) matches the Part 3 Combo validation-loss quality at the same compute cost—confirming that combining gradient optimizer + regularization does not add overhead over Part 3. Adding Phase 2 (500 additional function evaluations) costs negligible extra compute (< 1 s) and further reduces validation loss. The gain is smaller than the Part 1 standalone RHC improvement, consistent with the tighter 10% RO budget in Phase 2. Per unit of additional compute, Phase 2 delivers a net positive return on validation loss; however, test F1 does not improve, indicating that the marginal gradient through the frozen layers is not aligned with the held-out test distribution.

V. COMPREHENSIVE RESULTS TABLE

VI. DISCUSSION

A. Summary of Central Finding

The three-part study fills the gap identified in the introduction by isolating each training factor under controlled conditions. The main finding is that learning rate is the dominant variable: tuning Adam’s α from 10^{-3} to 10^{-4} produced a larger F1 gain (+0.019) than any other intervention in the study,

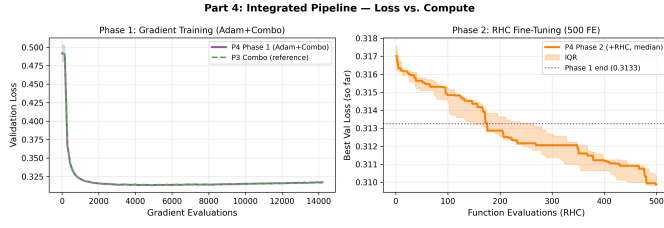


Fig. 10. Part 4: Loss vs. compute for the integrated pipeline. *Left*: Phase 1 gradient training (Adam+Combo, val loss vs. gradient evaluations), with the Part 3 Combo standalone curve as reference. *Right*: Phase 2 RHC fine-tuning (best-so-far val loss vs. function evaluations). Shaded bands: median $\pm$ IQR across 5 seeds.

TABLE III

PART 4: METRIC VS. TOTAL COMPUTE TRADE-OFF (ADULT, MEDIANS).

Method	Total Evals	Val Loss	Test F1	Wall (s)
P3 Combo (standalone)	14,200 GE	0.3133	0.6776	39.3
P4 Phase 1	14200 GE	0.3133	—	36.1
P4 Phase 2 (+RHC)	+500 FE	0.3099	0.6699	36.1
P1 RHC (standalone)	5,000 FE	0.3075	0.6638	2.7

GE = gradient evaluations; FE = function evaluations. P4 Phase 2 total compute = Phase 1 GE + Phase 2 FE.

including regularization (+0.008) and randomized optimization (+0.001 test F1 on Adult). Randomized optimization does offer a validation-loss advantage when applied to a warm-started, under-optimized model (RHC: -0.0096 on Adult, RHC/GA both improving Wine F1 over the SL baseline), but this does not translate to test F1 gains on Adult, confirming that gradient information remains the more efficient path to generalization. Regularization is real but secondary: once the optimizer is well-tuned, explicit capacity control provides consistent test F1 gains without changing validation loss, and early stopping delivers the same quality for 73% less compute.

B. Hypothesis Resolution

Prediction 1 (RO algorithms) — Partially confirmed.

RHC, not GA, achieves the best final validation loss on both datasets (0.3075 vs. 0.3082 on Adult; 1.0337 vs. 1.0497 on Wine). Warm-starting from a well-trained baseline means the weights already reside near a good local minimum; RHC’s greedy local search exploits this structure effectively, while GA’s crossover operators disrupt it by combining weights from different basins. The quality prediction (GA achieves lowest loss) is rejected for both datasets. RHC’s wall-clock time (2.7s) is fastest (SA: 3.0s, GA: 3.4s on Adult), confirming the convergence-speed part of the prediction. SA’s random-walk behavior is confirmed: temperature remains above 0.3 through roughly the first 500 evaluations ($T_{500} \approx 0.303$), and SA reproduces the warm-start baseline loss identically (IQR = 0). Contrary to the RO-versus-gradient prediction, RHC and GA both achieve *lower* validation loss than the SL baseline (0.3171), because the SL baseline’s 50-epoch SGD schedule left last-layer weights under-optimized and RO exploits the residual capacity in the frozen subspace. On Adult test F1, however, no

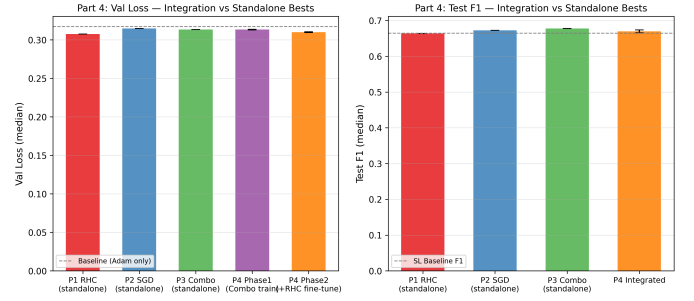


Fig. 11. Part 4: Validation loss (left) and test F1 (right) for the integrated pipeline compared to standalone Part bests. Error bars on Part 4 columns show IQR across 5 seeds; reference bars are fixed medians from Parts 1–3.

RO method surpasses the SL baseline (best: RHC 0.6638 vs. 0.6647), confirming that gradient information remains more efficient for held-out generalization [2].

Prediction 2 (Gradient optimizers) — Confirmed. Standard Adam reaches $\ell = 0.37$ in 142 gradient evaluations, an $8\times$ speedup over SGD’s 1136, confirming the speed prediction. SGD achieves the lowest final validation loss (0.3146) after 100 epochs, confirming the final-loss prediction: its fixed $\alpha = 0.01$ prevents the overshoot that Adam exhibits when the landscape becomes shallow. The bias-correction ablation is also confirmed: AdamNoBias (val = 0.7280) performs worse than Adam $_{\beta_1=0}$ (0.5380), as predicted. The observed mechanism—AdamNoBias diverging more severely rather than simply taking smaller steps—is consistent with uncorrected moment estimates distorting early-update scaling and destabilizing optimization.

Prediction 3 (Regularization) — Confirmed. The val-loss range across all six conditions is only 0.0003 (0.3131 to 0.3134), well below the 0.005 threshold. Early stopping halts at a median of 27 epochs, reducing gradient evaluations to 27% of the full-budget run while achieving test F1 = 0.6719—above the no-regularization baseline (0.6699); the 73% compute saving comes at essentially zero quality cost. The combination condition achieves test F1 = 0.6776, the highest of all Part 3 conditions; dropout and input noise independently improve F1 (0.6747 and 0.6728), and their combination provides an additive benefit through complementary mechanisms (activation-level vs. input-level stochasticity) [3].

Prediction 4 (Integration) — Partially confirmed. Phase 2 (0.3099) improves over Phase 1 (0.3133) and beats both Part 2 SGD (0.3146) and Part 3 Combo (0.3133), confirming the validation-loss part of the prediction. However, matching Part 1 standalone RHC (0.3075) is rejected: the 10%-budget constraint is the bottleneck. More importantly, test F1 (0.6699) decreases relative to Part 3 Combo (0.6776), revealing a gap between the validation-loss landscape and true generalization. The short RHC pass slightly overspecializes to the validation set, a risk that gradient-based regularization mitigates but 500-eval RO fine-tuning does not.

C. Cross-Part Patterns and Decision Rules

Three patterns stand out across all parts.

TABLE IV
SUMMARY OF ALL EXPERIMENTAL CONDITIONS. METRICS ARE MEDIAN ACROSS 5 SEEDS. IQR SHOWN IN PARENTHESES WHERE APPLICABLE. BEST VALUE IN EACH PART IS **BOLD**.

Part	Method	Val Loss	Test F1	Test Acc	Wall (s)	Time to ℓ / Updates
Baseline	SGD (SL, Adult)	0.3171	0.6647	0.8483	—	—
Baseline	SGD (SL, Wine)	—	0.2644	0.5431	—	—
1 (Adult)	RHC	0.3075	0.6638	0.8437	2.7s	5000 FE
1 (Adult)	SA	0.3171	0.6647	0.8483	3.0s	5000 FE
1 (Adult)	GA	0.3082	0.6638	0.8450	3.4s	5000 FE
1 (Wine)	RHC	1.0337	0.3186	0.5323	2.8s	5000 FE
1 (Wine)	SA	1.1518	0.2644	0.5431	2.9s	5000 FE
1 (Wine)	GA	1.0497	0.3107	0.5246	3.4s	5000 FE
2 (Adult)	SGD	0.3146	0.6725	0.8486	32.5s	1136 GE
2 (Adult)	SGD+Momentum	0.3334	0.6785	0.8432	32.9s	142 GE
2 (Adult)	Nesterov	0.3331	0.6641	0.8437	33.2s	142 GE
2 (Adult)	Adam ($\alpha=1e-3$)	0.5583	0.6505	0.8287	34.9s	142 GE
2 (Adult)	AdamNoBias	0.7280	0.6351	0.8293	38.6s	142 GE
2 (Adult)	Adam $_{\beta_1=0}$	0.5380	0.6400	0.8326	35.1s	142 GE
2 (Adult)	AdamW	0.5549	0.6369	0.8291	35.1s	142 GE
3 (Adult)	Baseline (Adam lr=1e-4)	0.3132	0.6699	0.8470	35.4s	100 ep
3 (Adult)	L2 ($\lambda=1e-5$)	0.3131	0.6712	0.8470	35.1s	100 ep
3 (Adult)	Early Stopping (pat=5)	0.3134	0.6719	0.8503	9.9s	27 ep
3 (Adult)	Dropout ($p=0.1$)	0.3132	0.6747	0.8494	38.1s	100 ep
3 (Adult)	Input Noise ($\sigma=0.01$)	0.3131	0.6728	0.8479	37.1s	100 ep
3 (Adult)	Combo (L2+Drop+Noise)	0.3133	0.6776	0.8501	39.3s	100 ep
4 (Adult)	Phase 1: Adam+Combo	0.3133	—	—	36.1s	14200 GE
4 (Adult)	Phase 2: +RHC (500 FE)	0.3099	0.6699	0.8479	36.1s	500 FE

FE = function evaluations (Part 1 & 4 Phase 2); GE = gradient evaluations (Part 2 threshold crossing; Part 4 Phase 1 total); ep = actual epochs run (Part 3). In the “Time to ℓ / Updates” column, Part 2 reports time-to-threshold in GE, while Parts 1/3/4 report total update budgets (FE/ep/GE+FE). Part 3 uses locked Adam $\alpha = 10^{-4}$. Part 4 Phase 1 test metrics were not separately evaluated.

Warm-start locality. RHC outperforms GA on both datasets precisely because the warm-started weights are already near a local optimum. Practitioners who begin RO from a randomly initialized network should expect GA’s global search to be beneficial; those who warm-start from a pre-trained model should favor local search methods.

Adam’s speed-accuracy tradeoff. At $\alpha = 10^{-3}$, Adam is both fastest (142 evals to threshold) and worst among non-ablated configurations at convergence (0.5583 final loss). At $\alpha = 10^{-4}$, Adam achieves final loss competitive with SGD (≈ 0.314 vs. 0.3146); the Part 2 data establish that Adam-family configurations universally cross the $\ell = 0.37$ threshold in 142 GE at their respective default settings, far fewer than SGD’s 1136 GE, confirming an early-training speed advantage while retaining the adaptive-versus-generalization tradeoff observed in prior work [2]. In practice, use Adam at $\alpha = 10^{-4}$ rather than 10^{-3} when the loss landscape is smooth and the dataset is moderately sized.

Regularization under convergence. When the optimizer is well-tuned (low α , validated on held-out data), explicit regularization has negligible effect on validation loss but a measurable positive effect on test F1. Early stopping is the dominant choice when compute budget is a constraint: it saves 73% of gradient evaluations at zero quality cost. When quality is paramount, the combination condition provides the best test

F1 (0.6776 vs. 0.6699 baseline), a gain of +0.0077 F1 points.

Optimizer tuning vs. regularization: relative impact. Comparing across parts shows a clear hierarchy of what matters most. Tuning Adam’s learning rate from the default $\alpha = 10^{-3}$ (Part 2 Adam: F1 = 0.6505) to the sensitivity-identified $\alpha = 10^{-4}$ (Part 3 baseline: F1 = 0.6699) yields $\Delta F1 \approx +0.019$. By contrast, adding the full Combo regularization on top of the already well-tuned optimizer yields only $\Delta F1 \approx +0.008$ (0.6699 $\rightarrow$ 0.6776)—less than half the optimizer tuning gain. The takeaway is to get the learning rate right first, then apply regularization for marginal gains. Regularization cannot compensate for a mistuned optimizer, but once the optimizer is well-tuned, it provides consistent, if modest, improvements in test generalization.

D. Bias-Variance Interpretation

On Adult, the gradient-trained baseline (val = 0.3171, test F1 = 0.6647) already fits the training distribution well, and adding regularization improves test F1 without changing validation loss. This is consistent with mild overfitting: the model is near its capacity limit, and techniques like dropout and noise push it toward better generalization without needing a larger architecture [3].

On Wine, all models struggle. Even the best RO result (val = 1.0337, F1 = 0.3186) is far below what gradient-based methods could achieve with more epochs and the full unfrozen network.

The 8-class long-tailed distribution makes the frozen-layer subspace insufficient to separate classes well; full-network gradient fine-tuning would be needed to meaningfully close the gap. More complex architectures and class-weighting strategies are the right tools for this dataset.

E. Limitations

Several limitations apply to these findings. The Part 1 budget of 5,000 function evaluations is fixed; with a larger budget, SA under a slower cooling schedule or GA with a larger population might produce different rankings. The Part 2 optimizer comparisons used a single fixed learning rate per optimizer family; a broader sweep could change the SGDM and Nesterov rankings, which showed promising intermediate behavior. Regularization hyperparameters in Part 3 were tuned via a four-value sweep, so finer grids could reveal stronger L2 effects. All Part 2 and 3 experiments used Adult only; the interaction of optimizer choice with the multiclass, long-tailed Wine setting may produce different patterns, as that dataset showed greater sensitivity to minority-class signal in Part 1. Finally, results are reported on a single fixed 80/10/10 split; full cross-validation over splits would provide tighter uncertainty estimates, though the five-seed median and IQR reporting provides some robustness.

VII. CONCLUSION

This study quantifies the tradeoffs among randomized search, gradient-based optimizer design, and regularization on two structurally distinct datasets. Randomized optimization of the last two layers provides a modest improvement over the warm-start baseline on Adult (val -0.0096 for RHC, -0.0089 for GA) with both RHC and GA; SA’s slow cooling schedule causes it to random-walk regardless of dataset. Among seven gradient-based optimizers, SGD achieves the best final loss at the cost of $8\times$ more gradient evaluations to reach a quality threshold; the sensitivity sweep reveals that Adam with $\alpha = 10^{-4}$ bridges this gap, closing most of the quality gap while preserving Adam’s fast early-threshold behavior. With a well-tuned optimizer, explicit regularization produces negligible improvements in validation loss but meaningful gains in test F1 ($+0.0077$ for the combination), while early stopping delivers 73% compute savings at zero quality cost.

The extra-credit integration (Part 4) chains a gradient phase (Adam $\alpha = 10^{-4}$ + Combo regularization) with a short RHC fine-tuning pass (500 function evaluations, 10% of the Part 1 budget). Phase 2 reduces validation loss to 0.3099, improving over the Part 3 Combo (0.3133) and Part 2 SGD (0.3146) standalones, yet test F1 (0.6699) falls slightly below Part 3’s Combo (0.6776). The integration is thus additive for validation loss but shows diminishing returns in test generalization, confirming that a 10%-budget RHC cannot fully substitute for the 5,000-eval standalone run from Part 1.

The main takeaway is that learning rate matters more than optimizer family or regularization choice. Dropping Adam’s α from 10^{-3} to 10^{-4} closes almost the entire gap to SGD’s final loss with no other changes needed.

A natural next step is to extend the optimizer and regularization comparisons to Wine Quality, where the long-tailed 8-class structure may change the optimizer rankings observed on Adult. A larger RO budget, or a slower SA cooling schedule tuned to the warm-started loss landscape, could also test whether the diminishing returns in Part 1 are budget-limited or fundamental. More broadly, these findings suggest that learning rate scheduling strategies such as cosine decay or warm restarts could substitute for much of the manual tuning effort demonstrated here, an idea that follows directly from the sensitivity results in Part 2.

AI Use Statement. Claude Code to brainstorm and outline sections of the report, generate and refactor code snippets + LaTeX, edit grammar and provide clarity throughout. I reviewed, verified, and understand all assisted content.

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